ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

ANALOGUE ELECTRONICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 2nd year Analogue Electronics paper 1 · practice a.

Written by Academic Central for practice, following analogue electronics modules at South African universities (e.g. University of Pretoria ENE 310 Analogue Electronics: amplifier concepts, op-amp circuits, oscillators and data conversion). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept answers that differ only through rounding, and the exact (Thevenin) bias method in QUESTION 2 if it is done correctly.

QUESTION 1

1.1.1 V_peak(secondary) = 12 × √2 = 16.97 V ✓ (3)
Two diodes conduct: 16.97 − 2 × 0.7 ✓ = 15.57 V ✓

1.1.2 Ripple frequency = 2 × 50 = 100 Hz ✓ (3)
$V_r = I / (fC) = 0.1 / (100 \times 2\,200 \times 10^{-6})$ ✓ = 0.45 V ✓

1.1.3 V_DC ≈ V_peak − V_r/2 = 15.57 − 0.23 ✓ = 15.34 V ✓ (2)

1.1.4 PIV ≈ V_peak(secondary) − 0.7 ✓ = 16.27 V (accept 16.97 V) ✓ (2)

1.2 I_S = (15 − 5.1) / 220 = 45 mA ✓✓ (6)
I_L = 5.1 / 510 = 10 mA ✓
I_Z = 45 − 10 = 35 mA ✓
P_Z = 5.1 × 0.035 = 0.18 W ✓✓

1.3 Sketch: positive half-cycles only, with gaps (zero) during the negative half-cycles ✓✓ (4)
Any TWO: the ripple frequency is doubled (100 Hz), so smoothing is easier with a smaller capacitor; both half-cycles are used, so it is more efficient; a higher average DC voltage; no DC current in the transformer secondary ✓✓

[20]

QUESTION 2

2.1.1 V_B = 12 × 10 / (47 + 10) ✓ = 2.11 V ✓ (2)

2.1.2 V_E = 2.105 − 0.7 = 1.41 V ✓ I_E = 1.405 / 1 000 = 1.41 mA ✓ (2)

2.1.3 I_C ≈ I_E = 1.41 mA ✓ (1)

2.1.4 V_C = 12 − 1.405 × 2.2 ✓ = 8.91 V ✓ (3)
V_CE = 8.91 − 1.41 = 7.50 V ✓

2.1.5 Yes: V_CE (7.50 V) is well above about 0.2 V and below V_CC, so the transistor is neither saturated nor cut off ✓ (1)

2.2.1 r_e = 25 mV / 1.405 mA = 17.79 Ω ✓ (3)
A_v = −R_C / r_e = −2 200 / 17.79 ✓ = −123.66 ✓

2.2.2 A_v = −R_C / (r_e + R_E) ✓ = −2 200 / 1 017.79 ✓ = −2.16 ✓ (3)

2.2.3 With CE the gain is high but depends on r_e, which changes with temperature and current ✓. Without CE the gain is low but almost fixed by R_C/R_E, so it is stable and less distorted (negative feedback) ✓ (2)

2.3 βr_e = 100 × 17.79 = 1.78 kΩ ✓ (3)
Z_in = R1 ‖ R2 ‖ βr_e = 47k ‖ 10k ‖ 1.78k ✓ = 1.46 kΩ ✓

2.4 I_C(sat) = V_CC / (R_C + R_E) = 12 / 3 200 ✓ = 3.75 mA ✓ (3)
V_CE(cut-off) = V_CC = 12 V ✓

2.5 They pass the AC signal ✓ but block DC, so the source and the load do not upset the bias point ✓ (2)

[25]

QUESTION 3

3.1 $A_v = 1 + Rf/R1 = 1 + 47/4.7$ ✓ $= 11$ ✓
$V_{out} = 11 \times 0.3 = 3.3$ V ✓ (3)

3.2 $V_{out} = -Rf (V_1/R + V_2/R + V_3/R)$ ✓ $= -(1 + 2 - 0.5)$ ✓ $= -2.5$ V ✓ (3)

3.3.1 $V_{out} = (100/10)(2.05 - 2.00)$ ✓✓ $= 0.5$ V ✓ (3)

3.3.2 CMRR is the ratio of the differential gain to the common-mode gain (often in dB) ✓. Both bridge outputs sit at about 2 V (a large common-mode voltage) while the useful signal is only 50 mV, so a high CMRR is needed to reject the 2 V and noise that is common to both wires ✓ (2)

3.4 BW = GBW / closed-loop gain = 1 MHz / 11 ✓ = 90.91 kHz ✓ (2)

3.5 $f_{max} = SR / (2\pi V_p)$ ✓ $= 0.5 \times 10^6 / (2\pi \times 10)$ ✓ $= 7.96$ kHz ✓ (3)

3.6.1 $f_c = 1 / (2\pi RC) = 1 / (2\pi \times 10\,000 \times 15 \times 10^{-9})$ ✓ $= 1.06$ kHz ✓ (2)

3.6.2 −3 dB at f_c ✓ about −20 dB at 10 f_c (the gain falls at 20 dB per decade) ✓ (2)

3.7.1 $\beta = R1/(R1 + R2) = 10/50 = 0.2$ ✓
Thresholds = ±0.2 × 12 = +2.4 V and −2.4 V ✓
Hysteresis = 4.8 V ✓ (3)

3.7.2 Its two thresholds (hysteresis) stop the output from switching many times when a slowly changing, noisy signal crosses one level ✓, giving one clean switch per crossing ✓ (2)

[25]

QUESTION 4

4.1 20 + 30 = 50 dB ✓
10^(50/20) ✓ = 316.23 ✓ (3)

4.2 $v_{in} = 10 \times 2\,000 / 2\,500 = 8$ mV ✓
Open-circuit output = 50 × 8 = 400 mV ✓
$v_{out} = 400 \times 400 / 500 = 320$ mV ✓
Overall gain = 320 / 10 = 32 ✓ (4)

4.3.1 $A_f = A / (1 + A\beta) = 10\,000 / 101$ ✓ $= 99.01$ ✓ (2)

4.3.2 $A_f = 5\,000 / 51 = 98.04$ ✓
A 50% drop in A changes the closed-loop gain by only about 1%: negative feedback makes the gain stable ✓ (2)

4.4 Class A: 25% (50% with transformer coupling) ✓ Class B: 78.5% ✓
THD: the ratio of the RMS value of all the harmonics in the output to the RMS value of the fundamental ✓, usually given as a percentage ✓ (4)

[15]

QUESTION 5

5.1 The loop gain Aβ must have a magnitude of 1 ✓ and a total phase shift of 0° (or 360°) around the loop ✓ (2)

5.2 $f = 1 / (2\pi RC) = 1 / (2\pi \times 10^{4} \times 10^{-8})$ ✓ = 1.59 kHz ✓
Gain of the amplifier = 3 ✓ (3)

5.3.1 Resolution = $5 / 2^{10} = 5 / 1\,024$ ✓ = 4.88 mV ✓ (2)

5.3.2 Code = 3.2 / 5 × 1 024 = 655.36 ✓, so 655 ✓
655 = 10 1000 1111 in binary ✓ (3)

5.4 $f_s \geq 2 \times 4$ kHz = 8 kHz (in practice higher) ✓
Aliasing: when a signal is sampled too slowly, its frequencies above $f_s/2$ appear as false, lower frequencies in the samples ✓✓ (3)

5.5 1010 = 10 ✓ $V_{out} = 8 \times 10/16 = 5$ V ✓ (2)

[15]

TOTAL: 100

END OF MARKING GUIDELINE